RAK5801 WisBlock 4-20mA Interface Module Datasheet

Overview

Description

The RAK5801 WisBlock Interface module was designed to be part of a production-ready IoT solution in a modular way and must be combined with a WisBlock Core and a Base module.

The RAK5801 is a 4-20 mA current loop extension module that allows the users to make an IoT solution for analog sensors with a 4-20 mA interface. This module converts the 4-20 mA current signal into voltage range supported by the WisBlock Core module (MCU) for further digitalization and data transmission.

The RAK5801 module features two input channels of 4-20 mA. Inside, a high-precision operational amplifier is used for signal amplification and conversion and supports a wide range of operating temperatures.

This module integrates a 12 V power supply, which can be used to power external sensors. The RAK5801 can be connected to 2-wire, 3-wire, or 4-wire types of 4-20 mA sensor. The module external interface is reached by a fast crimping terminal that allows connection for the 4-20 mA sensors (including power) and to the I2C bus. The fast crimping terminals can be used without the need of special tools, which simplifies the installation process on the field.

Features

- Two 4-20 mA analog inputs
- Compatible with multiple WisBlock Core modules, such as RAK4631
- 0.1 mA conversion accuracy
- Supports low power consumption mode. The module can be powered off by the WisBlock Core module for saving energy during idle periods.

- 12 V output to power external sensors
- Reserved I2C expansion interface
- Fast crimping terminals
- Designed with a 2 kV ESD protection level
- Chipset: STMicroelectronics LM2902
- Small dimensions of 35 mm x 25 mm

Specifications

Overview

The overview discusses the block diagram of the board. It also shows the installation mechanism on how to mount the board into the baseboard.

Block Diagram

The RAK5801 module was designed to convert 4-20 mA current signals into voltage signals by applying a sampling resistor. As shown in **Figure 1**, the input current signal from the sensor is conditioned by an operational amplifier to match the level supported by the ADC input of an MCU where the signal is digitized.

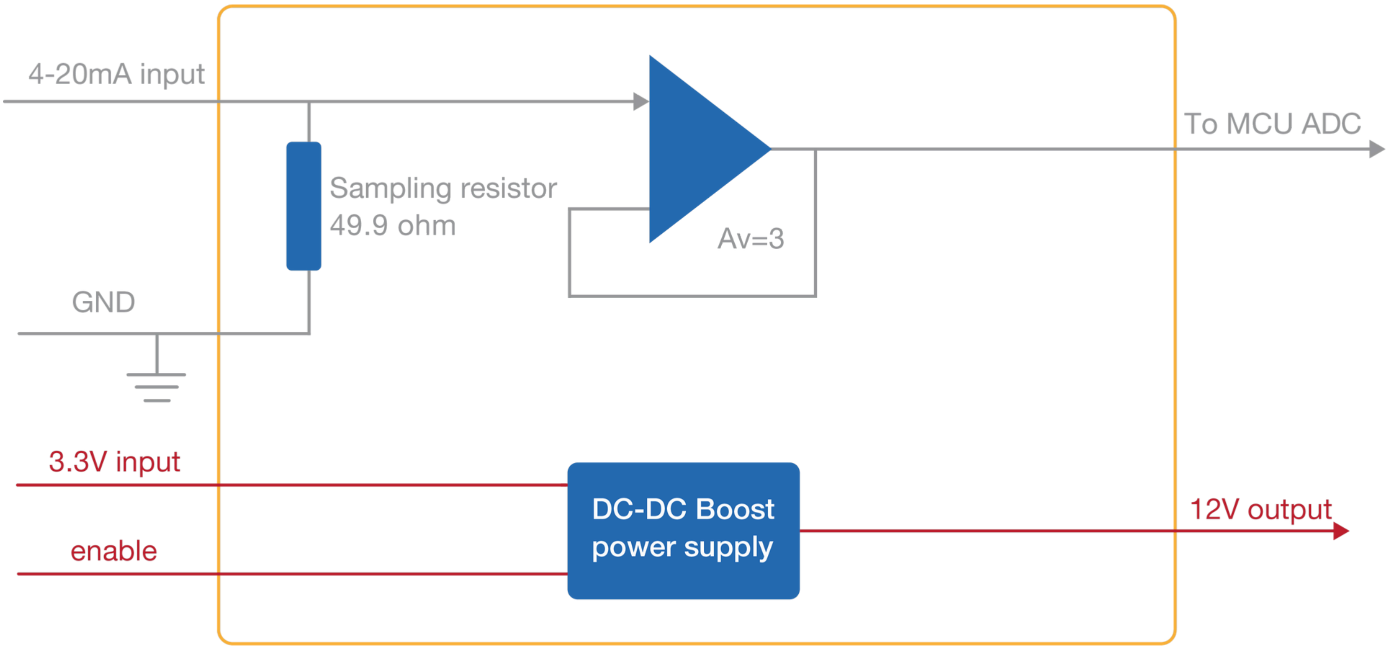


Figure 1: RAK5801 Block Diagram

Once the signal is digitalized, you can recover the original current value by applying the following formula:

```
I = U/149
```

Where **U** is the ADC reading and **I** the sensor current.

As shown in **Figure 1**, the module provides an output of 12 V for powering passive 4-20 mA sensors. This 12 V output is boosted by an internal DC-DC booster. The enable pin allows to control the power conversion module and set the RAK5801 module into a low power consumption mode.

Hardware

The hardware specification is categorized into four parts. It discusses the pinouts of the board and its functionalities and diagrams.

Chipset

Vendor	Part number
STMicroelectronics	LM2902

Device Specification

The following table shows the parameters and the description of the RAK5801 WisBlock 4-20 mA Interface Module:

Parameter	Description
Analog Input Interface	2 channels of 4-20 mA
Analog Sampling Resolution	0.005 mA
Analog Sampling Accuracy	1%

Parameter	Description
Analog Maximum Input Current	25 mA (There is a risk to burn the circuit surpassing this limit.)
Analog Port ESD Protection Level	2 kV HBM
Current Sampling Resistor	49.9 Ω
Operational Amplifier Gain	3.0
Input Voltage	3.0-3.6 V
Output Voltage	12 V
Output Current	Maximum 30 mA
Operating Temperature	-30 °C ~ 65 °C
Storage Temperature	-40 °C ~ 85 °C
Module Dimensions	35×25 mm

Pin Definition

This section covers the pin number of the sensor connector, the definition, and the functionalities of each pin shown in a tabular representation.

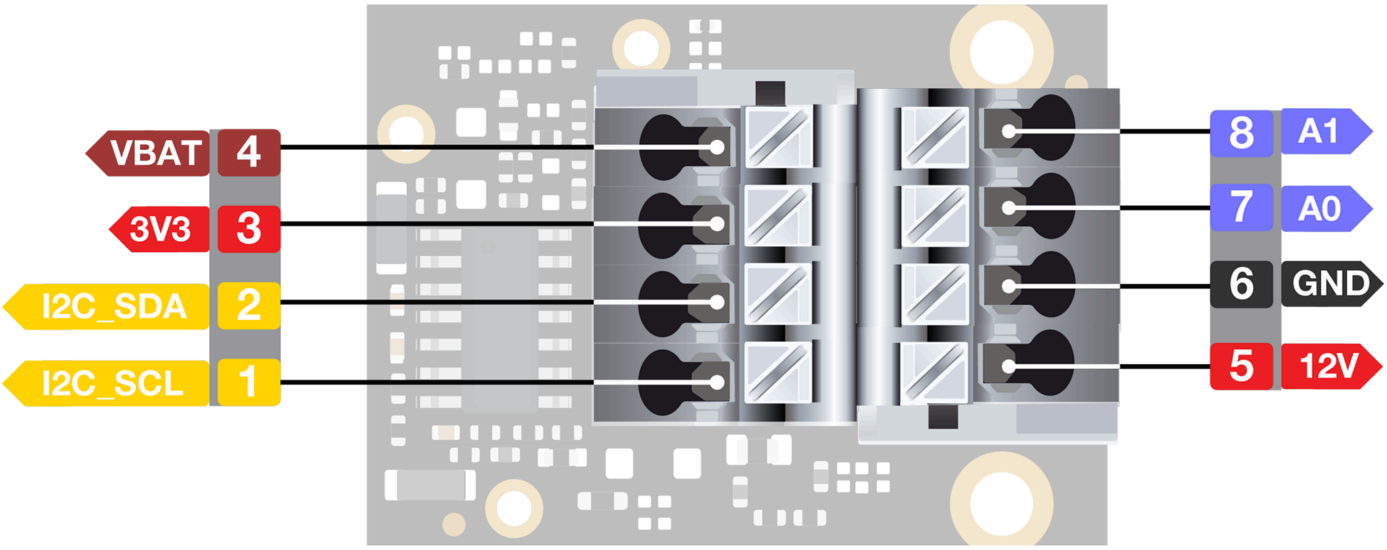


Figure 2: RAK5801 Sensor Connector

> **NOTE**
>
> **A0** cannot be used as an analog input channel by default because it is used for measuring battery voltage. But if you need to use **A0**, there are few hardware modifications needed to configure.
>
> To enable **A0** as an additional channel:
>
> 1. Remove **R7** on the WisBlock Base such as the RAK5005-O to disconnect Vbat sensing.
> 2. On RAK5801, remove the 0 Ohm resistor in **R94** and put it to **R95**. See **Figure 3**.

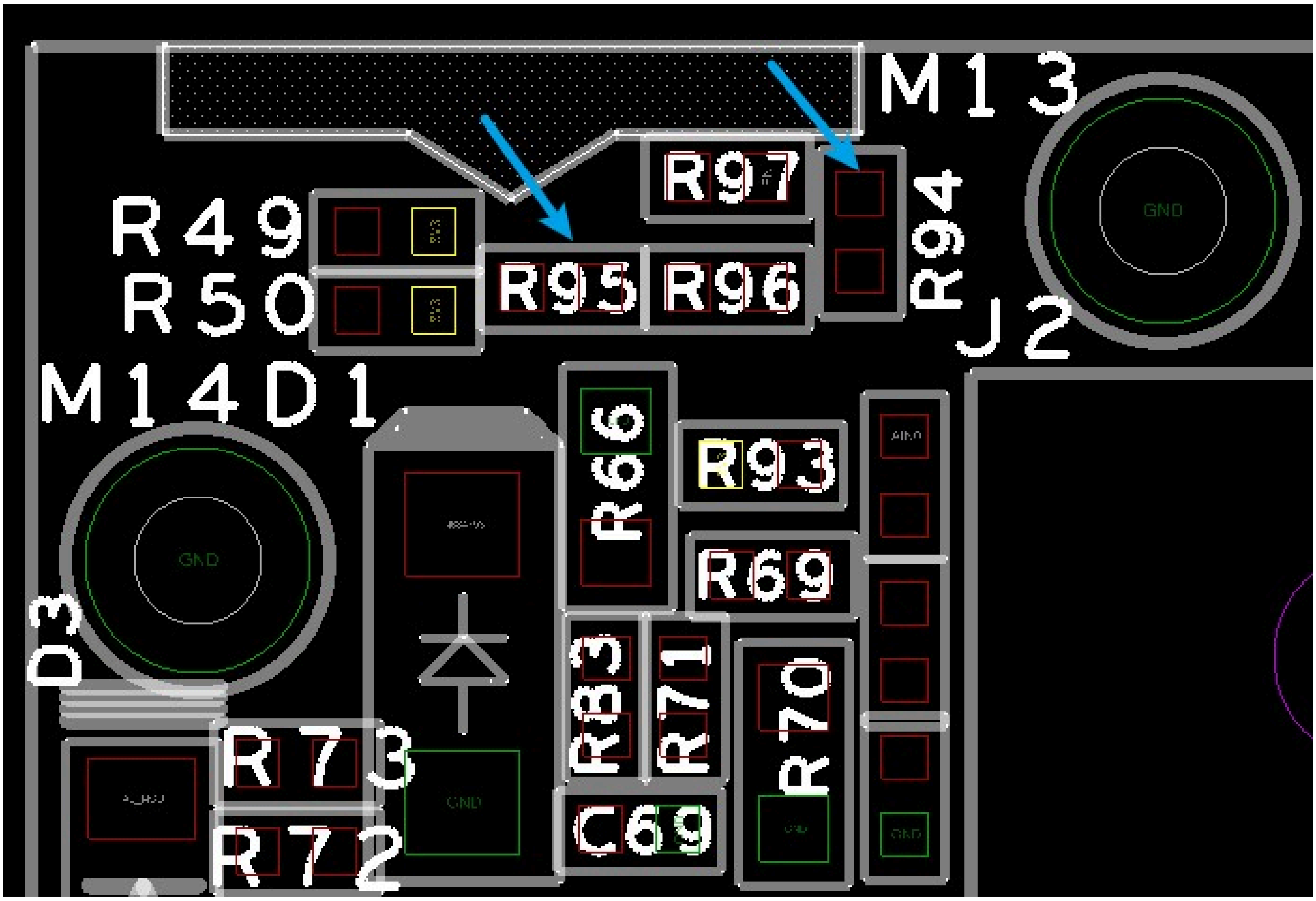


Figure 3: A0 Hardware Modifications

Pin Number	Function Description
1	SCL of the I2C interface
2	SDA of the I2C interface
3	3V3 output
4	VBAT, Battery output
5	12 V output for external sensors
6	GND
7	Analog input 0

Pin Number	Function Description
8	Analog input 1

Figure 4 shows the pin order for the IO connector of the module. Through this connector, the RAK5801 module is attached to the WisBoard baseboard.

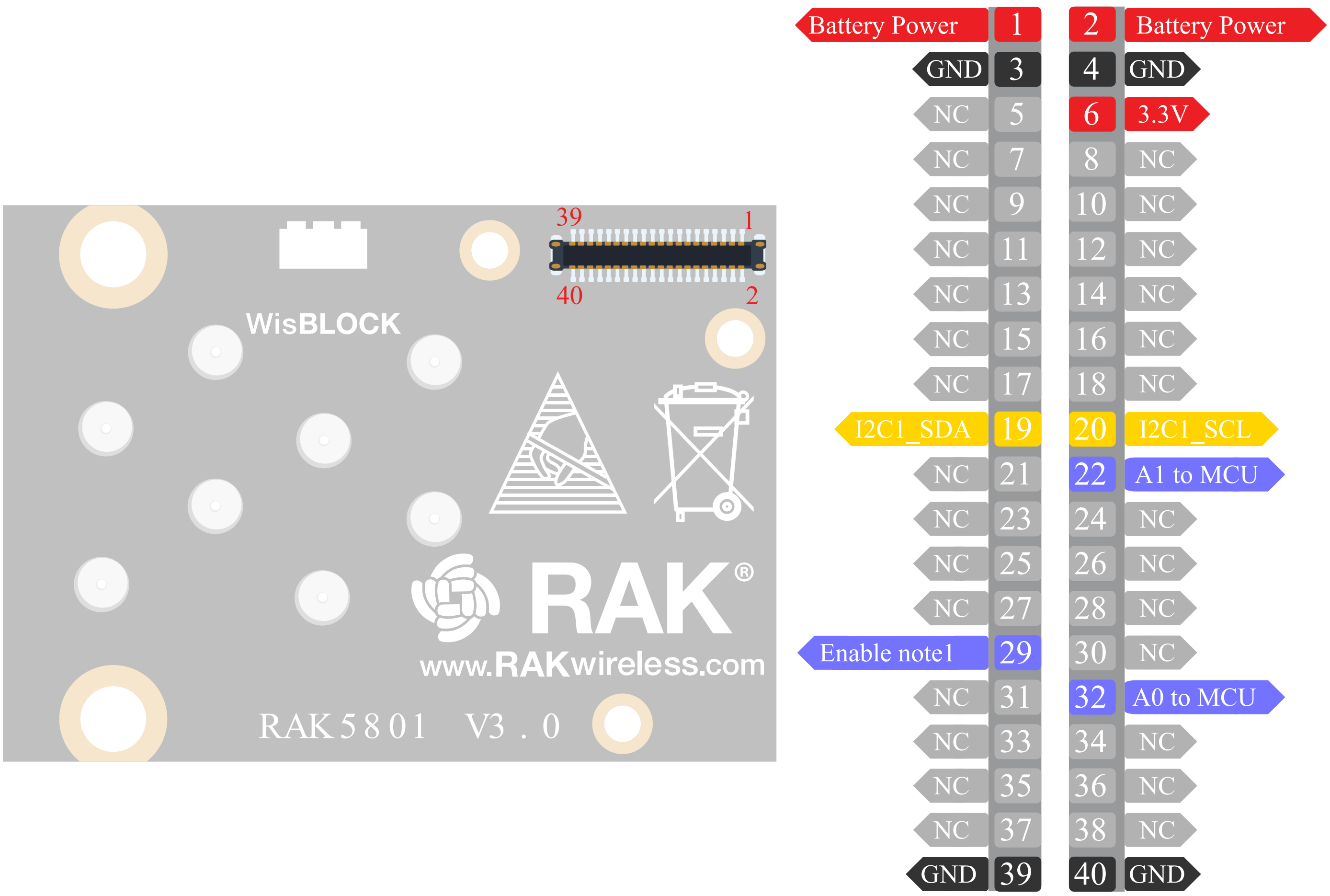


Figure 4: RAK5801 Internal WisIO Connector

The functionalities of each pin of the WisIO connector are tabulated below:

Pin Number	Description	Pin Number	Description
1	Battery Power	2	Battery Power
3	GND	4	GND

Pin Number	Description	Pin Number	Description
5	NC, reserved for 3V3	6	3.3V Power
7	NC	8	NC
9	NC	10	NC
11	NC	12	NC
13	NC	14	NC
15	NC	16	NC
17	NC	18	NC
19	SDA for I2C1	20	SCL for I2C1
21	NC	22	Analog to MCU
23	NC	24	NC
25	NC	26	NC
27	NC	28	NC
29	Enable note1	30	NC
31	NC	32	Analog0 to MCU
33	NC	34	NC
35	NC	36	NC

Pin Number	Description	Pin Number	Description
37	NC	38	NC
39	GND	40	GND

NOTE

This signal controls the dc-dc power supply on RAK5801, before capturing the analog signal, set this pin to high to enable power for RAK5801.

Mechanical Characteristics

Board Dimensions

Refer to **Figure 5** below for the mechanical dimensions of the RAK5801 module.

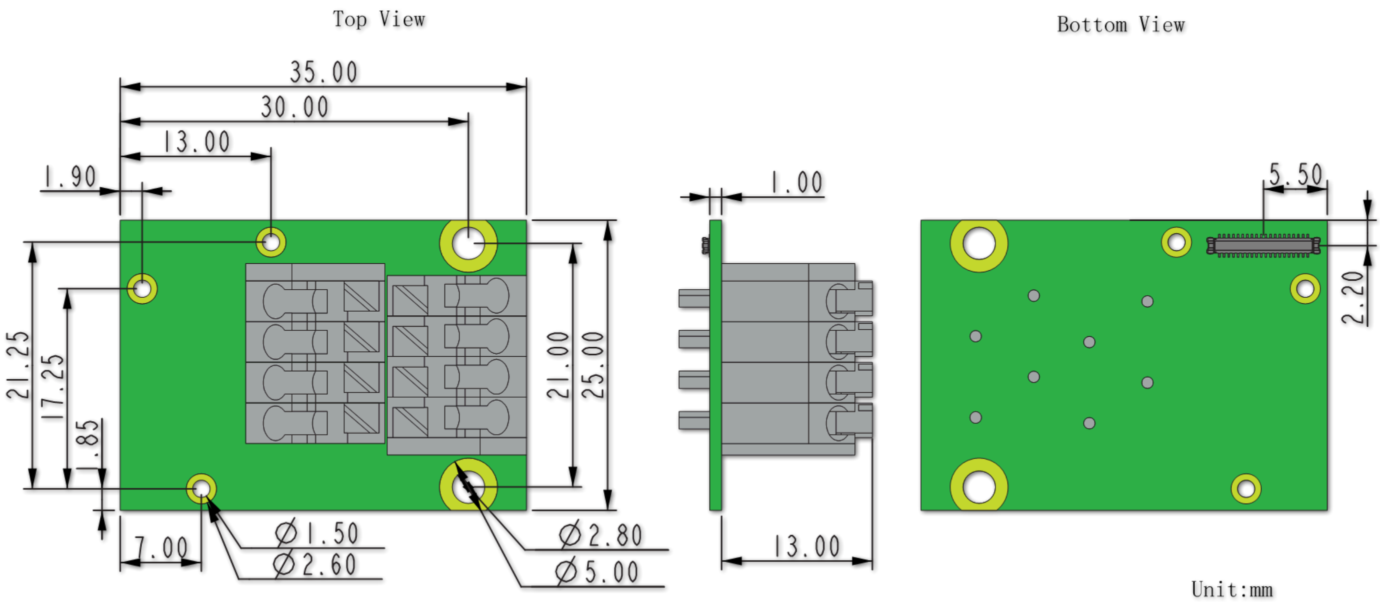
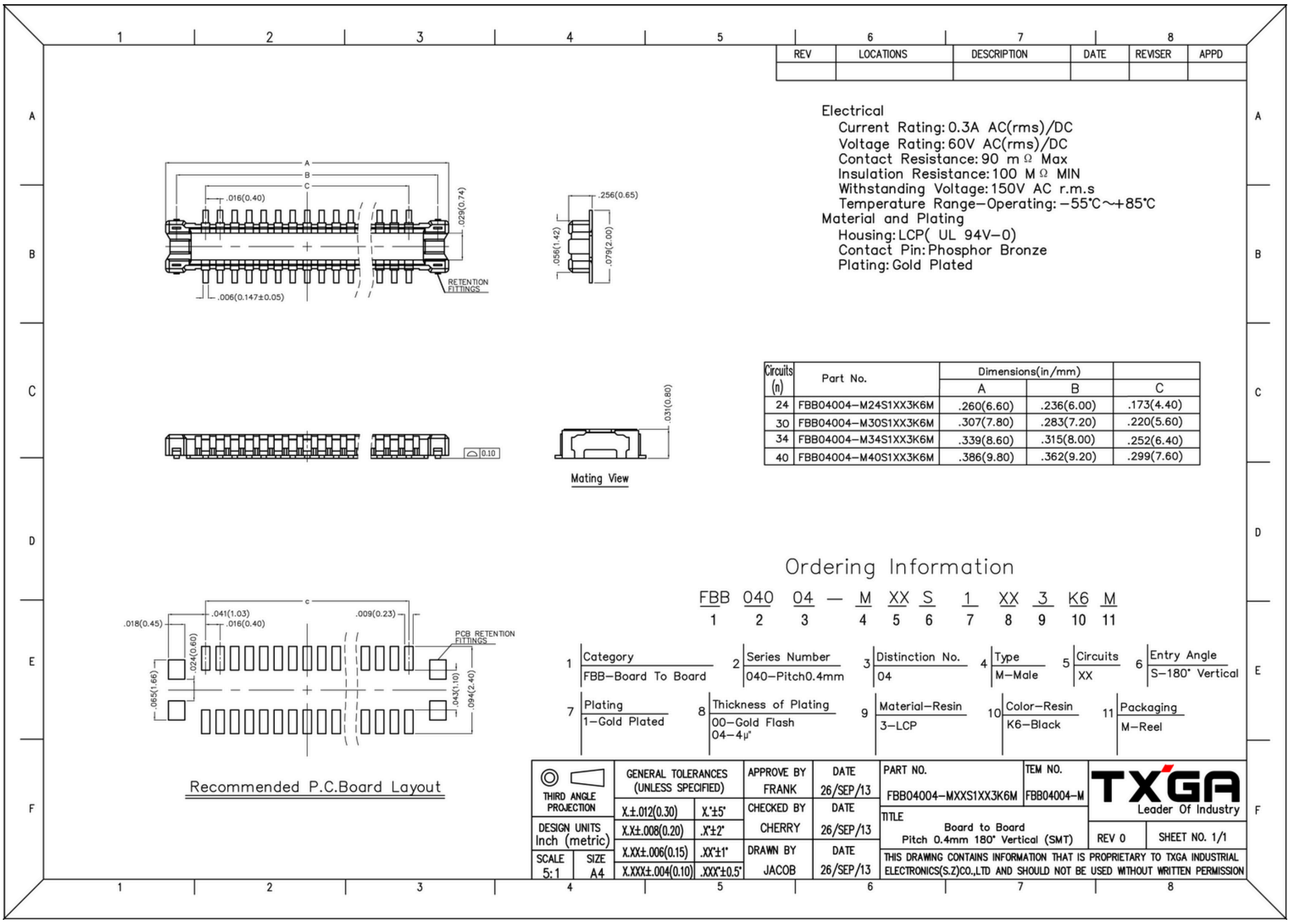


Figure 5: RAK5801 Mechanical Dimensions

WisConnector PCB Layout



Circuits (n)	Part No.	Dimensions(in/mm) A	B	C
24	FBB04004-M24S1XX3K6M	.260(6.60)	.236(6.00)	.173(4.40)
30	FBB04004-M30S1XX3K6M	.307(7.80)	.283(7.20)	.220(5.60)
34	FBB04004-M34S1XX3K6M	.339(8.60)	.315(8.00)	.252(6.40)
40	FBB04004-M40S1XX3K6M	.386(9.80)	.362(9.20)	.299(7.60)

Figure 6: WisConnector PCB footprint and recommendations

Schematic Diagram

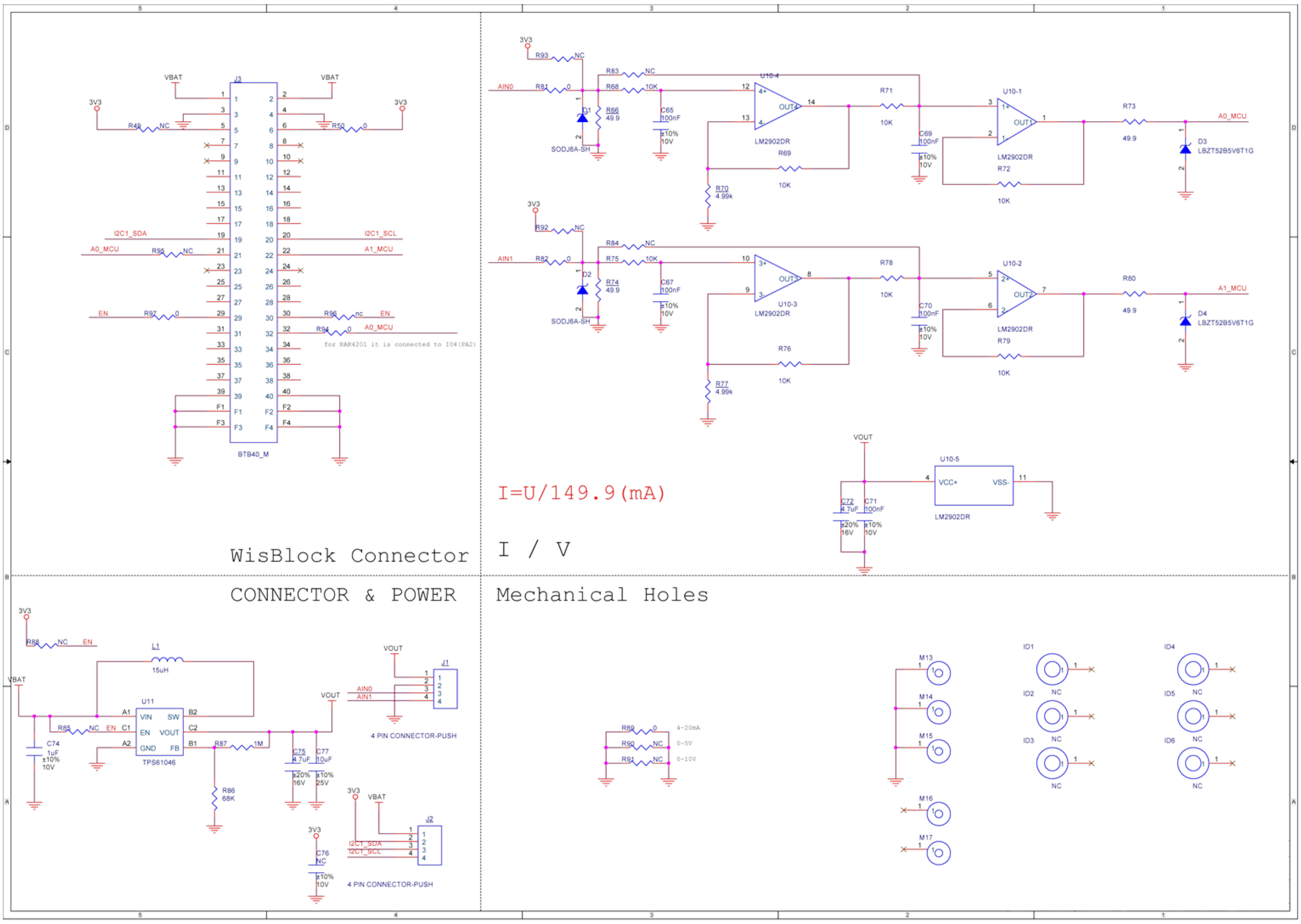


Figure 7: RAK5801 Schematic Diagram

WisBlock Compatibility

Since a WisBlock module can be combined with a variety of different functional modules, the pin functions of the MCU are multiplexed, so the interface expansion module for each specific function may need to be properly adapted for the WisBlock. The compatibility details of the RAK5801 module are as shown in the table below:

WisBlock Module	Adaptable Module	Description
WisBase Base board	RAK5005/RAK5005-O	RAK5801 is designed to be assembled in the IO slot of the RAK5005-O baseboard.

WisBlock Module	Adaptable Module	Description
WisBlock Core Module	RAK4631	RAK5801 is compatible with RAK4631.
	RAK4201	Select RAK4201L-ADC for the low band or RAK4201H-ADC for the high band.
	RAK4202	Refer to Note 2 for hardware adaptions to the RAK5005-O and RAK5801.
	RAK4261	Refer to Note 3 for hardware adaptions to the RAK5005-O and RAK5801.

NOTE

1. The RAK5801+RAK4601 The RAK5801 is not compatible with RAK4601. The main reason is that RAK4601 doesn't expose any ADC pin through the RAK5005-O baseboard.

NOTE

2. RAK5801+RAK4202+RAK5005-O
In order to combine a RAK5801 module, a RAK4202 (WisBlock Core module), and the RAK5005-O, the following modification must be introduced:

- In RAK5005-O, remove the R7 resistor, as shown in **Figure 8**.
- In RAK5801, remove R94 to R95 resistors. Use **PA0 of STM32L151** to read the analog data of the channel "analog0" and **PA2 of STM32L151** to read the analog data of Channel analog1. **Figure 9** shows the resistors R94 and R95 on the RAK5801 module.

This combination has the following restriction:

- The adapted RAK5005-O module will not be able to sense the battery voltage anymore.

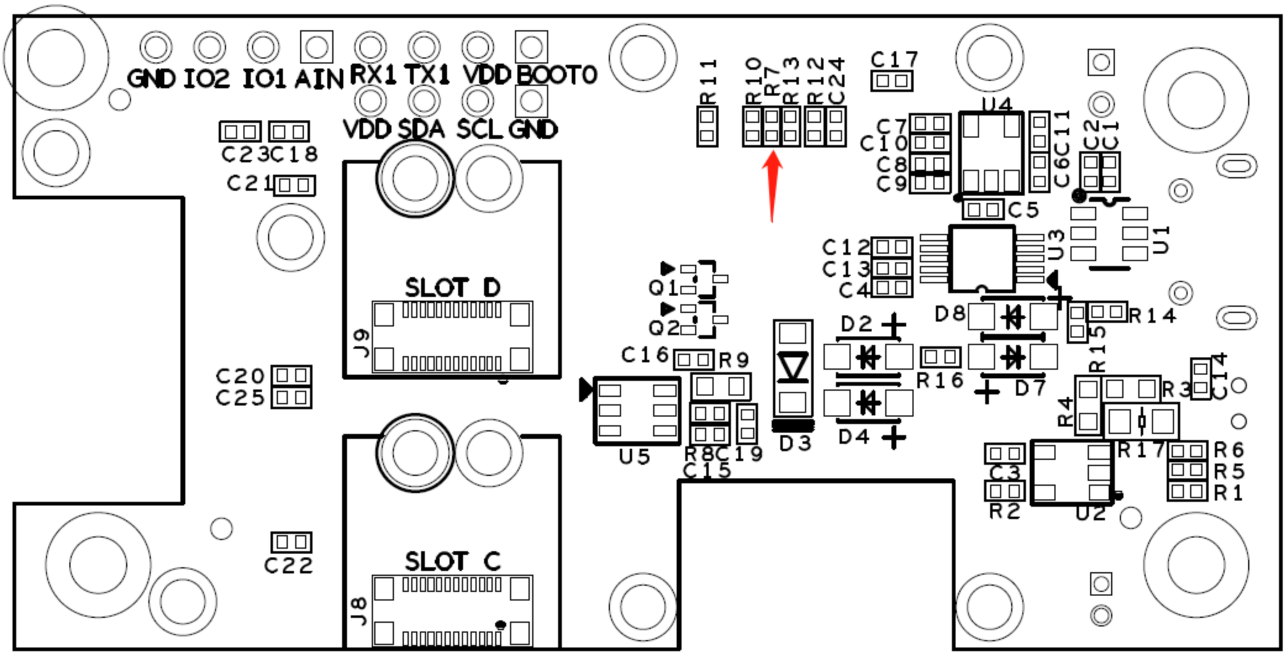


Figure 8: R7 on RAK5005-O

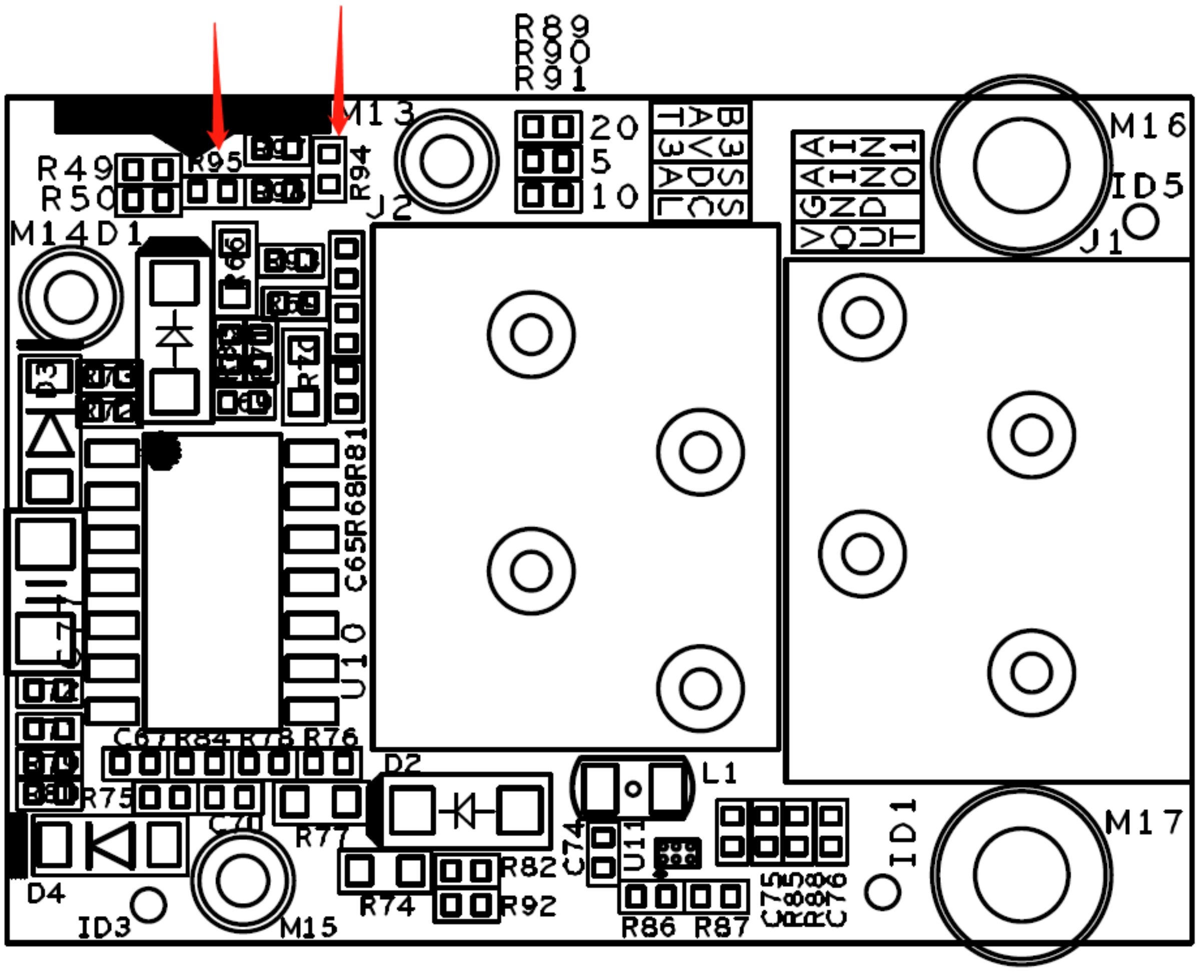


Figure 9: R94 and R95 on RAK5801

NOTE

3. RAK5801+RAK4261+RAK5005-O

In order to combine a RAK5801 module, a RAK4261(WisBlock Core module), and the RAK5005-O, the following modification must be introduced:

- In RAK5005-O, remove the R7 resistor. See **Figure 8**.
- In RAK5801, remove R94 to R95 resistors (see **Figure 9**). Use **PA08 of ATSAMR34** to read the analog data of the channel "analog0" and **PA09 of ATSAMR34** to read the analog data of Channel analog1.

This combination has the following restriction:

- The adapted RAK5005-O module will not be able to sense the battery voltage anymore.